

Script of the Design Course

Design Course: RF- and Microwave Circuits and Systems

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Version 12.0



EEH, Fritz-Hüttinger chair energy-efficient
high-frequency electronics
Freiburg University

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Introduction

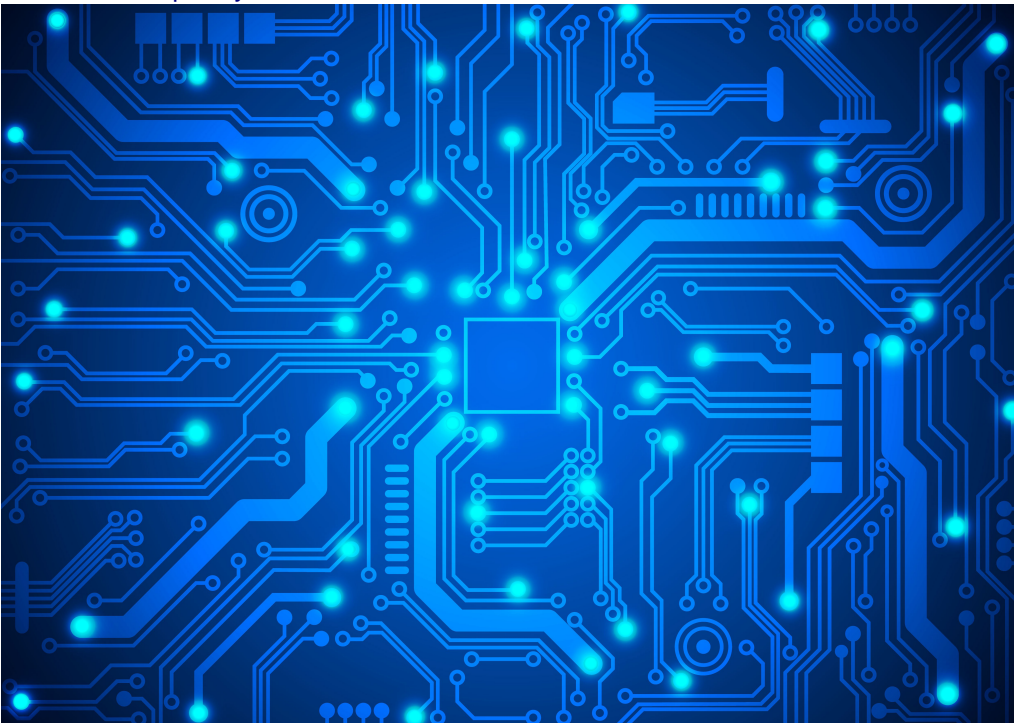


Introduction



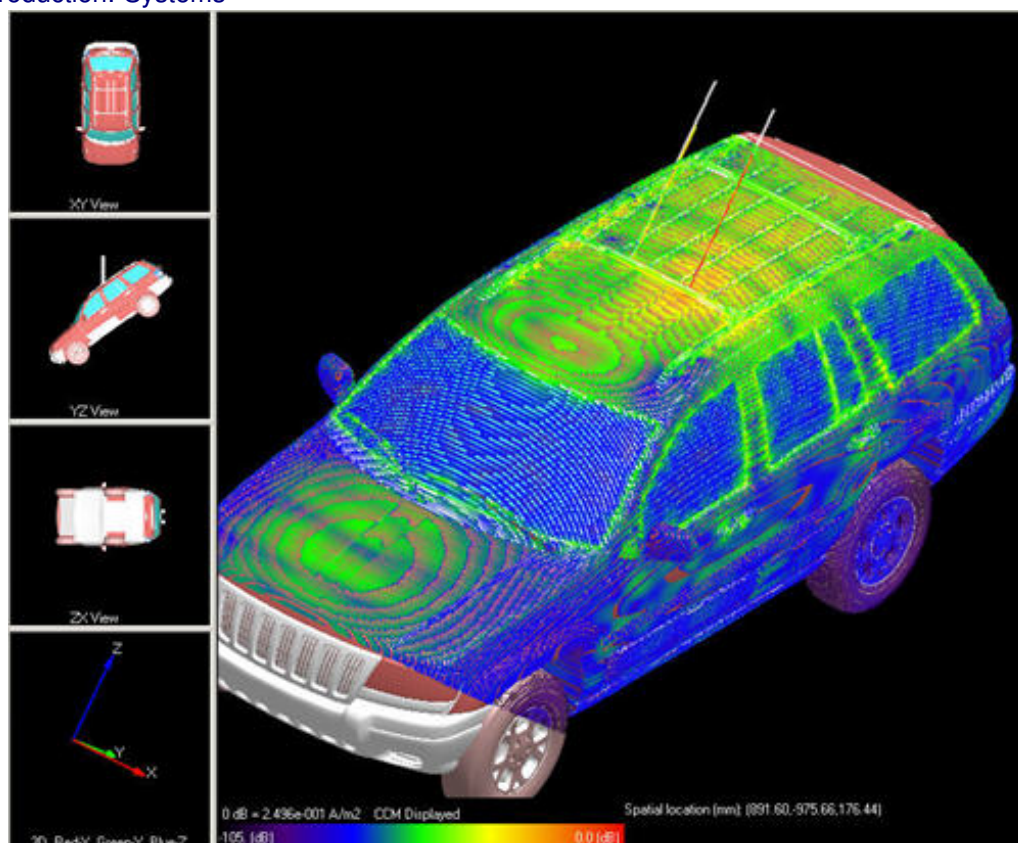
1.4

Introduction: Complexity



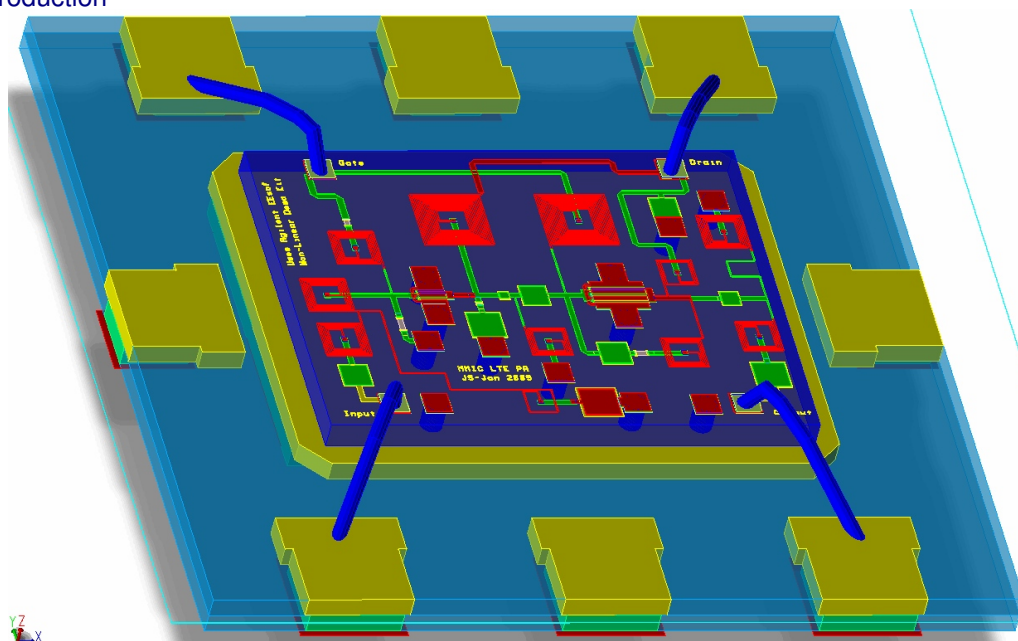
1.5

Introduction: Systems



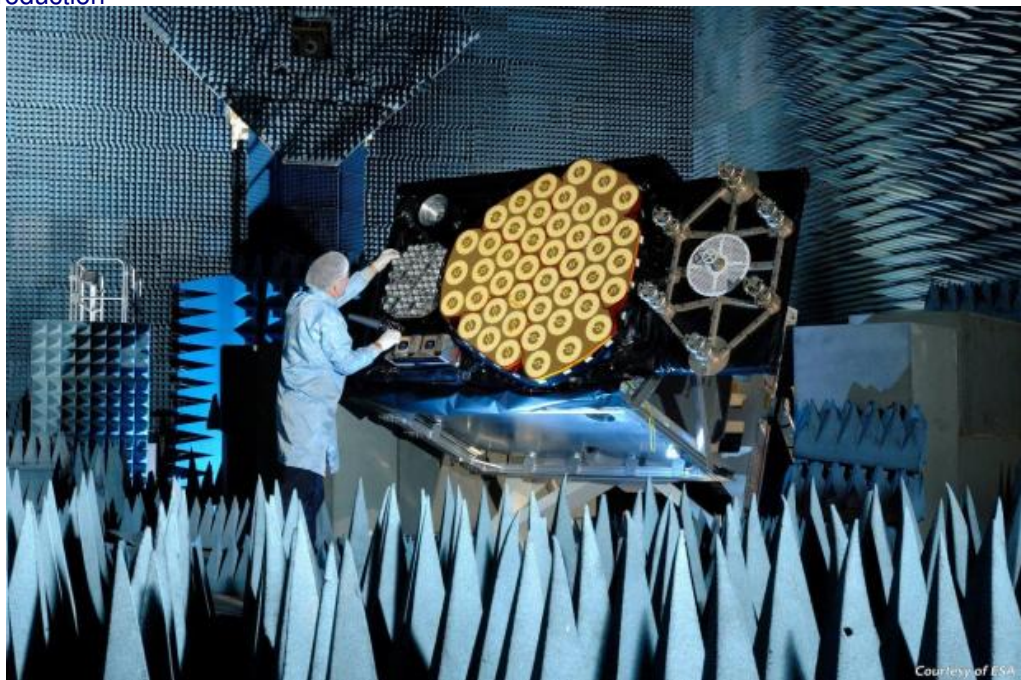
1.6

Introduction



!!!! Frequency domain !!!! vs. Cadence Virtuoso: Time domain

1.7



Dates with exercise

- Mo 1.) 20.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 2.) 27.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 3.) 4.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 4.) 11.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 5. 18.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 6. 25.05.2026 no lecture, pentacost break SR 00 014 (G.-Köhler-Allee 078)
- Mo 7. 1.06.2026 live SR 00 014 (G.-Köhler-Allee 078)

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Dates with exercise II

- Mo 8. 8.6.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 9. 15.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 10. 22.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 11. 29.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 6.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 13.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 13. 20.07.2026 Presentation of final results from competition 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)

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Date and Location

Dates with exercise

1. Mo, 20.4.2026: 16-17.00 h Introduction and group formation
2. Block 1: Mo., 27.4.2026: (issue block 1a on ILIAS)
3. Block 1: Mo., 4.5.2026: (issue block 1b on ILIAS)
4. Block 2: Mo., 11.5.2026 (issue block 2a on ILIAS), (Deadl. Block 1)
5. Block 2: Mo., 18.5.2026 (issue block 2b on ILIAS)
6. Block 3: Mo., 1.6.2026 (deadline block 2 on ILIAS), (issue block 3a on ILIAS)
7. Block 3: Mo., 08.6.2026 (issue block 3b on ILIAS)
8. Block 4: Mo., 15.6.2026 (Deadline block 3), (issue block 4a on ILIAS)
9. Block 4: Mo., 22.6.2026, (Issue Block 4b, COMPETITION !!!)
10. Block 5: Mo. 29.6.2026 (deadline block 4), (issue block 5a on ILIAS)
11. Block 5: Mo., 06.7.2026 : (issue block 5b ON ILIAS)
12. Mo., 13.7.2026, Final presentations based on 4b (deadline block 5 on ILIAS)

1.11

Rules of the Exercises

- 5 blocks Exercises (every one/two weeks) with overall max. 10 tasks in a given format...
- First: starting 27.4.2026... (two weeks to do, sometimes more)
- 4 blocks to be completed successfully (with 50% correct) (three people per group)
- Voluntary Competition: Final presentation (5 mins) of the group to the plenary based on a task
- Exercises are given out on ILIAS and also submitted there.
- **Timing is day of the deadline, 16 h**

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1 Abstract

The students will be enabled to understand, design and layout modern RF- and microwave components and systems by means of the electronic design environment Advanced Design System including the two- and three dimensional electromagnetic simulators Momentum and EMPro 3D. The detailed use of a complex RF-software environment is a dedicated target of this course. This includes the numerical analysis of complex passive and active devices, the design and layout of hybrid and integrated circuits, and their packaging and signal flow. The students will be competent to design and layout passive and active RF-structures including packages and interconnects and circuits of relevance to everyday communication and sensing. The competence includes in-depth understanding and treatment of complex microwave systems and of general system design including the treatment of complex modulated signal flows.

Introduction

Materials and Script

The material is provided at the beginning and during the lecture.

Email: ruediger.quay@inatech.uni-freiburg.de

- Foils: Intro materials from Keysight
- ADS Video Library on www.youtube.com
- Script (supporting materials)
- Exercises

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Videos

1. Watch ADS videos on youtube using ADS : www.youtube.com , Keyword: ADS Video Library
2. Basics and setup:
3. <https://www.youtube.com/watch?v=04BfajzXnmM&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz>
4. Understanding Workspaces:
5. <https://www.youtube.com/watch?v=7jNm400BaiE&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=2>
6. The basics of schematics:
7. <https://www.youtube.com/watch?v=wqyyOCzxGS0&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=4>
8. Data displays:
9. <https://www.youtube.com/watch?v=jIg10RihLQk&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=6>

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WWW Support 1

ILIAS Support

<https://ilias.uni-freiburg.de>

search for: Design Course: RF- and Microwave Systems

password: design2026

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WWW Support 2

This exercise is recorded, attendance is, however, required



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Windows Access to the Uni Network

- Get login at university, get additional TF login
- Install VPN from University server
- use ssh from linux
- If you start from Windows:
- Use, e.g., xming as windows X-server (for export of the display)
- Install Putty ssh client on windows

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Form groups of Three

- The exercises are submitted from a group of three people
- Find two partners !!!
- Send your group to ruediger.quay@inatech.uni-freiburg.de, including TF login name.
- Exercises start april 27th 2026 !

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Install an ADS Version on a local computer

Last Updated: 05 March 2020

ADS - Windows® Operating Systems	
Product	Windows
ADS 2020 / ADS 2020 Update 1.0 / ADS 2020 Update 2.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2019/ ADS 2019 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2017/ ADS 2017 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2016.01 64-bit Operating Systems	Windows 10 (64-bit)* Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2015.01 64-bit Operating Systems	Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)

You can work on the aragorn but local has proven to be easier...

Organize your data in a good structure

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Login to Application Server for the license

From Remote: Create RAS Access in Mylogin

Go into University net with VPN: check VPN Uni Freiburg

In case you use Windows to do that: use putty and install xming as windows X-server

Application Server: `aragorn.imtek.uni-freiburg.de`, `ssh-X-1 (yourlogin) aragorn.imtek.uni-freiburg.de`

Shells usable: `bash` and `tcsh`

Start: `ads` after using `ssh` to the server

Licence Server will be asked: `27010(at)frodo.imtek.uni-freiburg.de`

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Backup Server !

Application Server: `samweis.imtek.uni-freiburg.de` (only via VPN)

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Recommendation

Use Licence server only, install ADS on your local device ! Use ADS 2023 Update 1.0

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The following references may be useful to provide more detailed information.

References

- List of useful ADS Videos
- Keysight Software Support: <http://www.keysight.com/>
- Introduction to ADS, ADS_Basics.pdf (ILIAS)
- Momentum Momentum_Basics.pdf (ILIAS)

1.23

2 Oscillator Integrated Circuits (ICs)

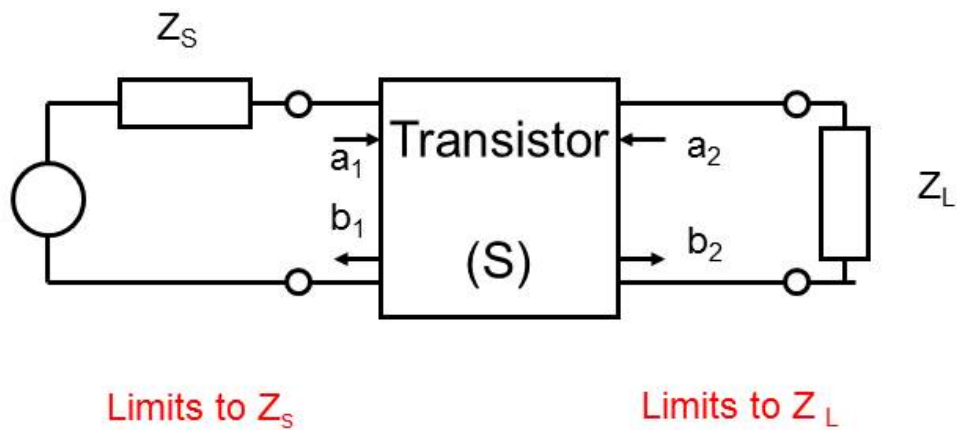
Educational Targets

- Understand how to use the undesired instability to your advantage.
- Understand the Smith Chart as a means to describe (instability) and reflection coefficients.
- Use varactors as tunable capacitors.
- Overcome the idea that only lumped inductors and capacitors can form resonances.
- Understand the built-in feedback at microwave frequencies !
- Understand importance of tuning to stability !
- Understand the phenomenon of phase noise and perform a phase noise simulation.
- Use for the first time a Design rule checker on a given layout.

2.1 Stability in Amplifier

Small-Signal Amplifier

Stability: a Big Topic with Various Instances



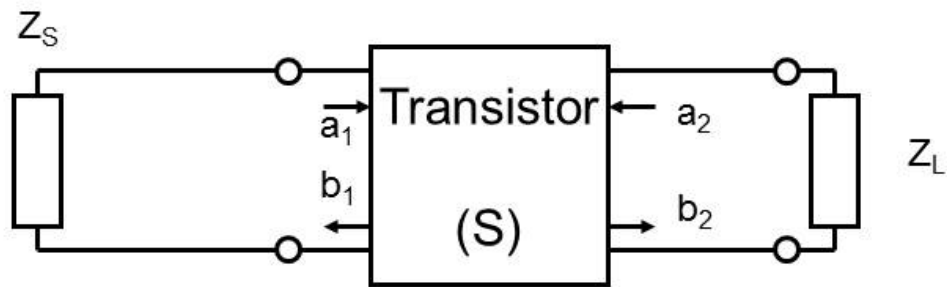
Amps: In-stability is annoying!

Make use of the otherwise "undesired" in-stability

2.2 De-Stabilization

Example: Oscillator

De-Stabilization



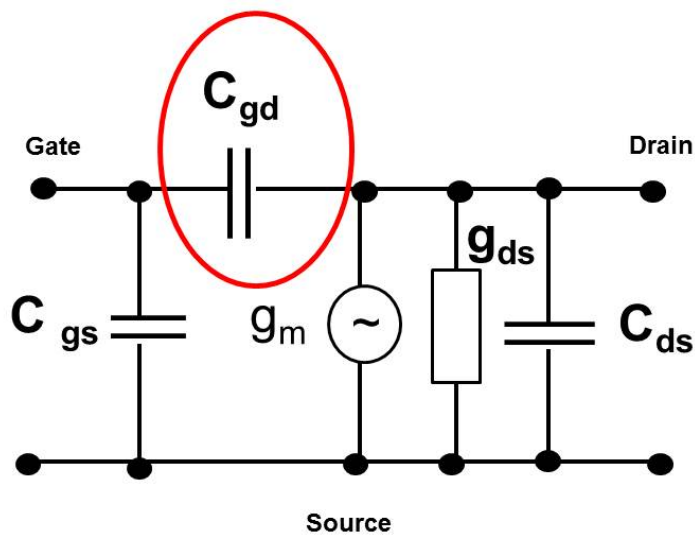
$$|\Gamma_1 \Gamma_S| > 1 @ f_{osc} \quad \text{or} \quad |\Gamma_2 \Gamma_L| > 1 @ f_{osc}$$

Destabilize

7.27

Oscillators from Amplifiers Through Internal Feedback

Stability



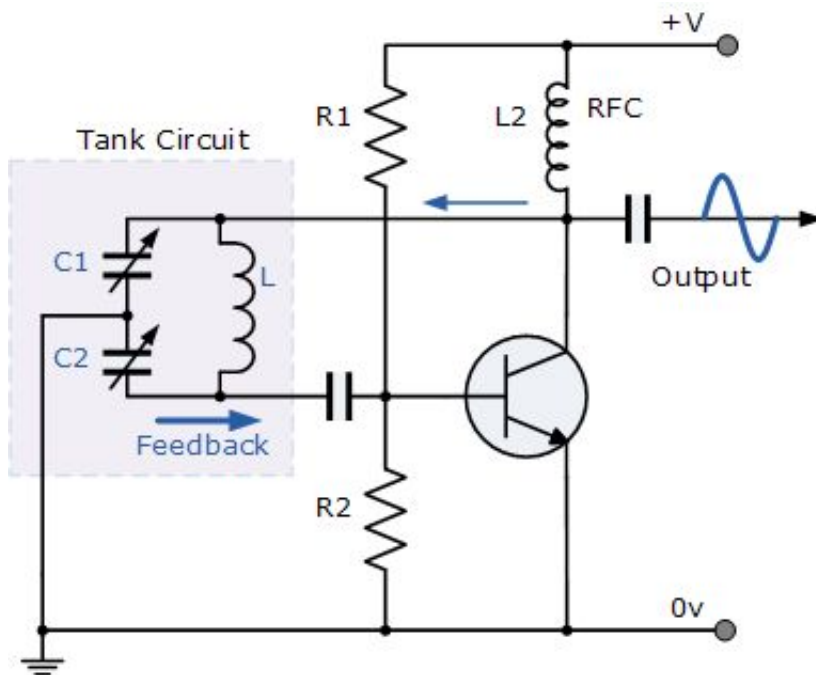
FET: Intrinsic feedback capacitance C_{gd} , Bipolar: C_{bc} or C_{jc} : both devices are not-unilateral

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2.3 Types of Oscillators

LC Oscillators: AC Concept

Colpitts Oscillator with Bipolar Transistor



Voltage divider from two capacitors C_1 , C_2 , one inductor in a tank

Concept to be migrated to microwave frequencies !

7.29

2.4 Phase Noise

Phase Noise

Ideal oscillator

$$u(t) = U_0 \cdot \cos(\omega_0 \cdot t) \quad (1)$$

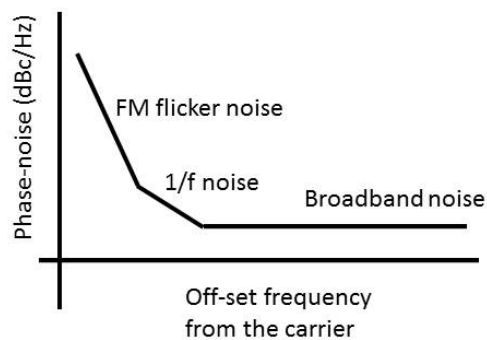
Real oscillator

$$u(t) = U(t) \cos(\phi(t)) = U(t) \cdot \cos(\omega_0 \cdot t + \Phi(t)) \quad (2)$$

Phase Noise term $\Phi(t)$: rapid, short-term, random fluctuations in the phase of a signal

7.30

Phase Noise II: Spectral Distribution



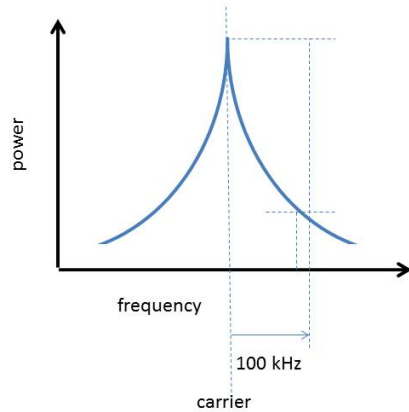
$$\text{Phase noise } L(f) = \frac{S_{\Phi}}{2} \text{ Units of dBc/Hz} \quad (3)$$

S_{Φ} : single-side phase noise spectral density

7.31

Phase Noise III

Spectrum: Phase noise is a design parameter



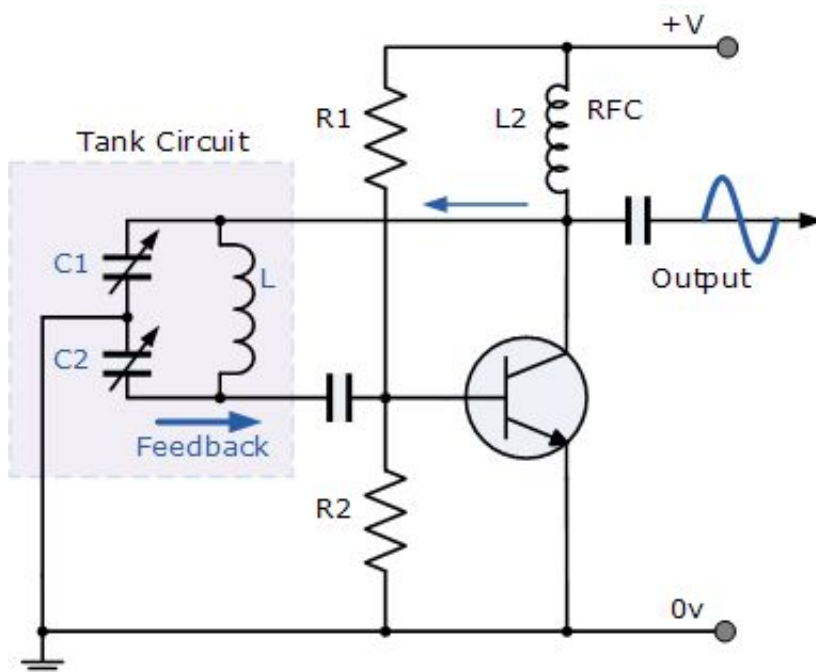
Phase noise ($L(f)$): units of dBc/Hz

7.32

2.5 Types of Oscillators

LC Oscillators: AC Concept

Colpitts Oscillator with Bipolar Transistor

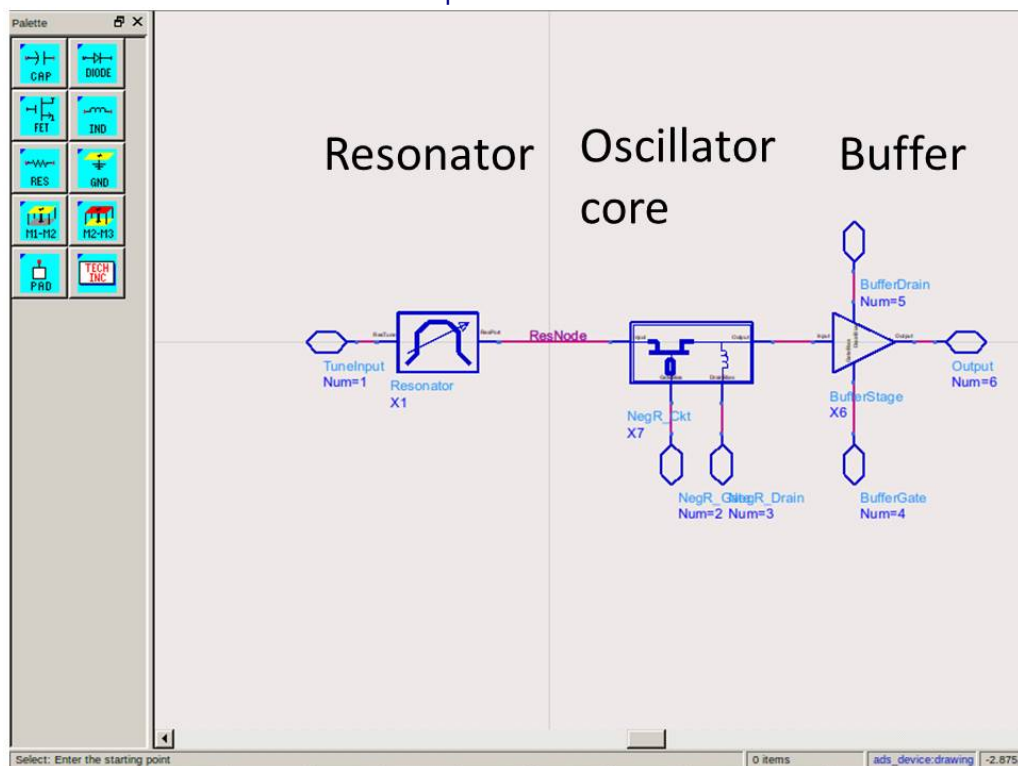


Voltage divider from two capacitors C_1 , C_2 , one inductor in a tank

Concept to be migrated to microwave frequencies !

7.33

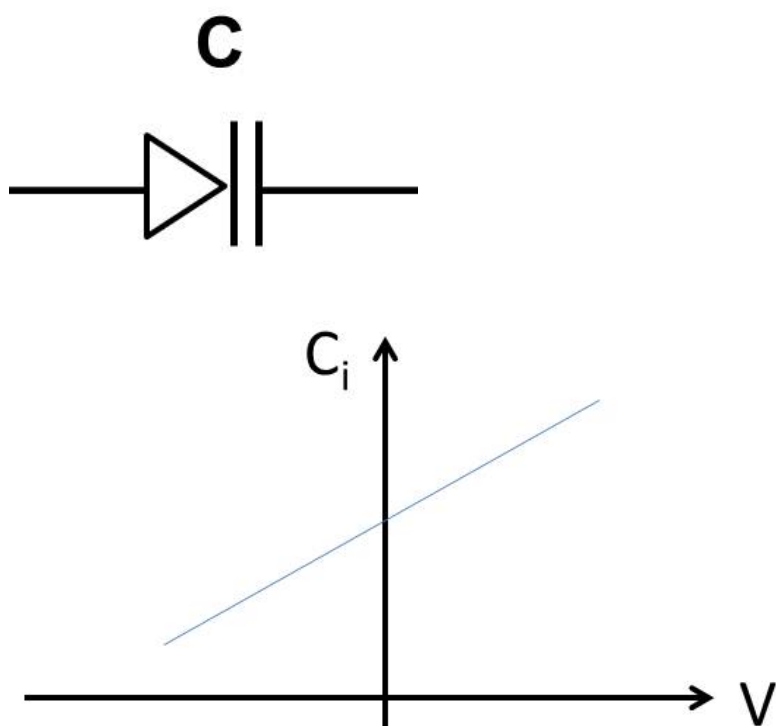
Microwave Oscillators: Microwave Concept



7.34

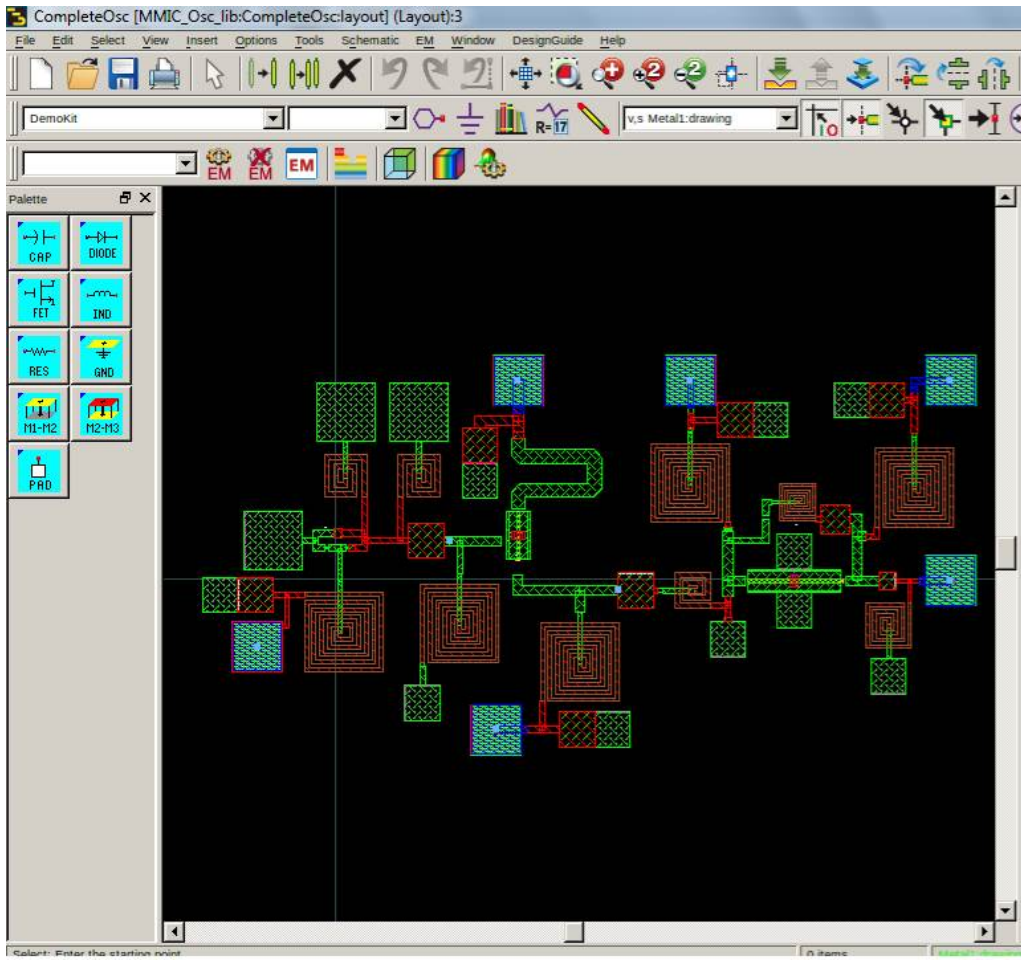
Tuning: Linear Varactor: Linear Variable Capacitor

Varactor: tuneable PN junction: tuneable capacitance linear in capacitance and voltage in resonator



7.35

First encounter with Layout and Design Rules Check



3 Tasks 11: Microwave Oscillator Circuit Design (Block 3 Part 2)

3.1 Tasks 11

Tasks 11: Microwave Oscillator Circuit Design (Block 3 Part 2)

1. Explanation: Open the example in MW CTS MMIC Osc lib
2. Task: In the example go to `Varactortest`. What is the main electrical representation of a varactor (three sentences)? What can be added in the description? (5 points)
3. Task: For the loop creation: How can $\gamma_x > 1$ be achieved electrically? (one plot and three sentences) (5 points)
4. Task: Go to the schematic of `CompleteOscZTBSchematic` and perform the simulation as default. Explain why the product of the $\gamma_x \times \gamma_y$ justifies an oscillation condition! Which two conditions need to be achieved and what is the frequency vs. phase relation! (in max. 5 sentences) (5 points)

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Tasks 11: IC Design Continued (Block 3 part 2)

5. Task: Goto to `NegRTestSimple`. Run the simulation for V_{dc} at source 1 = -0.31 V. Compare to the situation for V_{dc} at source 1 from -0.31 V to the situation at -0.9 V. Explain what happens at/near this bias (-0.9 V) with the transistor in terms of current vs. voltage and thus vs. stability. Think: What happens with the gain there (i.e. at pinch-off). Can a device in the off-state be unstable? (5 points)
6. Explanation: Go to the Schematic: `ResonatorTB`
7. Task: Dive into the resonator and explain (in five sentences) which function the varactor and its bias dependence has there? What is the defining quantity of a varactor in this case? (name one) Give its definition (one equation)! (5 points)
8. Explanation: Go to the Schematic: `CompleteOscwTLsEMTB` Task: Change the supply voltages of the oscillator core to $V_{dc} = 2.35$ V (not of the buffer). How much do you have to adjust the tuning voltage V_{bias} of the voltage-controlled oscillator to get back to 20 GHz? (one plot and one sentence) What happens to the output power of the oscillator at fundamental frequency as a function of bias? (one plot + one sentence) (5 points)

7.38

Tasks 11: IC Design Continued (Block 3 part 2)

9. Task: For $V_{dc} = 2.35$ V, perform the phase noise simulation. Apart from bias: Name two measures to minimize phase noise in an oscillator, e.g. 100 kHz off-set? (2 sentences) (5 points)
10. Task: Go to the schematic `BufferStage_wTLsSP_TB`! Explain why for the function of the buffer amp the S12 is checked! Which situation is thought of? (3 sentences) (5 points)

7.39

4 Tasks 12: IC Layout and Design Rule Check

4.1 Tasks 12: IC Layout and Design Rule Check

Tasks 12: IC Layout and Design Rule Check (Block 3 part 2)

1. Explanation: Go to `CompleteOsc` and look at the layout given.
2. Explanation: Go to Tools, go to DRC.
3. Task: Name examples of rule violations in a DRC and give typical related errors in an integrated circuit! Search the www, if you have no clue... (5 points)
4. Task: Go to Tools again: Run ADS Desktop LVS (layer vs. schematic). Starting from the schematic: Describe (max. 3 sentences) what this important tool LVS does for you? What can go wrong? (5 points)
5. Task: Try to create three critical changes in the layout and see what the LVS tells you for your own reference: Document your example (5 points)
6. Task: Try to remove a component to the design in the schematic (`CompleteOsc`), rebuild the layout, and run the LVS. Are you still error free? Document! (5 points)

7.40

Tasks 11 format for 40 points

1. 11.2 Give three sentences max...(5 points)
2. 11.3 Give the plot and three sentences ! (3 points)
3. 11.4 IN YOUR OWN WORDS: Give five sentences max. (2 points).
4. 11.5 Give the two 2 plots (5+5 points) and four sentences.
5. 11.7 Give five sentences, one name, and one definition (5 points)
6. 11.8 Give one plot and one sentence, again one plot and one sentence, max. (5 Points)
7. 11.9 Give one plot and two sentences (5 Points)
8. 11.10 Three sentences (5 Points)

7.41

Tasks 12 format for 20 points

1. 12.3 Give one table with five lines and a column with errors each... (5 points)
2. 12.4 Give three sentences max...(5 points)
3. 12.5 Give three sentences max. and one plot (5 points)
4. 12.6 Give three sentences max and max. one plot (5 points)

7.42